

In G. 3/08 (OJ 2011, 10) (Question 2 of the referral), the Enlarged Board of Appeal stated that a claim in the area of computer programs can avoid exclusion under Art. 52(2)(c) and (3) EPC merely by explicitly mentioning the use of a computer or a computer-readable storage medium. It noted however that no exposition of this position would be complete without the remark that it is also quite clear from the case law of the boards of appeal since T. 1173/97 that if a claim to program X falls under the exclusion of Art. 52(2) and (3) EPC, a claim which specifies no more than "Program X on a computer-readable storage medium," or "A method of operating a computer according to program X," will always still fail to be patentable for lack of an inventive step under Art. 52(1) and 56 EPC.

The Enlarged Board of Appeal also noted that in T. 424/03 (extending the reasoning applied in T. 258/03) the board held that claim to a program ("computer executable instructions" in the claim in question) on a computer-readable medium also necessarily avoided exclusion from patentability under Art. 52(2) EPC. The decision in T. 424/03 went on to note that the particular program involved had the potential of achieving a **further technical effect** when run and thus also contributed to the technical character of the claimed subject-matter. The Enlarged Board of Appeal stated that this fact was, however, not necessary to the conclusion that the claimed subject-matter avoided exclusion, since according to the reasoning of T. 258/03 **any technical means** claimed was **sufficient** to overcome the exclusion of Art. 52(2) EPC.

2.4.5 Patentability of computer programs: technical character

Whereas the view had been taken on a number of occasions in the boards' earlier case law (T. 26/86, OJ 1988, 19, T. 110/90, OJ 1994, 557, T. 164/92, OJ 1995, 305, and T. 204/93) that the exclusion from patentability under Art. 52(2)(c) and (3) EPC 1973 applied to all computer programs, independently of their contents, that is, independently of what the program could do or perform when loaded into an appropriate computer, the board in T. 1173/97 (OJ 1999, 609) held that, in the context of applying the EPC, the technical character of an invention was generally accepted as an essential requirement for its patentability, as was illustrated, for instance, by R. 27 and 29 EPC 1973. The exclusion from patentability of computer programs as such (Art. 52(2) and (3) EPC 1973) could be construed to mean that such programs were considered to be mere abstract creations, lacking in technical character. The use of the expression "shall not be regarded as inventions" seemed to confirm this interpretation. This meant that computer programs had to be considered patentable inventions **when they had a technical character**.

Further, Art. 27(1) TRIPS provides that "patents shall be available for any inventions, whether **products or processes**, in all fields of technology". The board went on to state that it was the clear intention of TRIPS not to exclude from patentability any inventions, whatever field of technology they belong to, and therefore, in particular, not to exclude programs for computers as mentioned in and excluded under Art. 52(2)(c) EPC 1973. The board held that a computer program product is not excluded from patentability under Art. 52(2) and (3) EPC 1973 if the program, when running on a computer or loaded into a computer, brings about, or is capable of bringing about, a **technical effect** which goes beyond the "normal" physical interactions between the program (software) and the computer (hardware) on which it is run.

The combination of the two provisions (Art. 52(2) and (3) EPC 1973) demonstrated that the legislators did not want to exclude from patentability all programs for computers. In other words, the fact that only patent applications relating to programs for computers as such were excluded from patentability (see T 208/84, OJ 1987, 14 and T 115/85, OJ 1990, 30) meant that patentability could be allowed for patent applications relating to programs for computers where the latter were not considered to be programs for computers as such.

2.4.6 Establishing technical character

a) Further effects of programs for computers

In T 1173/97 (OJ 1999, 609) and T 935/97 the board found that a computer program must be considered to be patentable when it has technical character. For the purpose of interpreting the exclusion from patentability of programs for computers under Art. 52(2) and (3) EPC 1973, the board assumed that **programs for computers** could not be considered as having a technical character for the very reason that they are programs for computers. This means that physical modifications of the hardware (causing, for instance, electrical currents) deriving from the execution of the instructions given by programs for computers cannot per se constitute the technical character required for avoiding the exclusion of those programs. Although such modifications may be considered to be technical, they are a common feature of all those programs for computers which have been made suitable for being run on a computer, and therefore cannot be used to distinguish programs for computers with a technical character from programs for computers as such. It was thus necessary to look elsewhere for technical character in the above sense: It could be found in the **further effects** deriving from the execution (by the hardware) of the instructions given by the computer program. The board stated that every computer program product produced an effect when the program concerned was made to run on a computer. The effect only showed in physical reality when the program was being run. Thus the computer program product itself did not directly disclose the said effect in physical reality. It only disclosed the effect when being run and consequently only possessed the "potential" to produce said effect. This effect might also be technical in the sense explained in reason 6 of the decision, in which case it constituted the "further technical effect" mentioned there. This meant that a computer program product might possess the potential to produce a "further" technical effect.

The board concluded that on condition they were able to produce a technical effect in the above sense, all computer programs had to be considered as inventions within the meaning of Art. 52(1) EPC 1973, and might be the subject-matter of a patent if the other requirements provided for by the EPC were satisfied.

b) Direct link to physical reality

The President of the EPO referred the following point of law to the Enlarged Board of Appeal (G 3/08, OJ 2011, 10; Question 3(a) of the referral): must a claimed feature cause a **technical effect on a physical entity** in the real world in order to contribute to the technical character of the claim? It was noted in the referral that according to decisions

T.163/85 and **T.190/94**, a technical effect **on a physical entity** in the **real world** was required. This was, however, not the case in **T.125/01** and **T.424/03**. In these decisions the technical effects were essentially confined to the respective computer programs.

The Enlarged Board of Appeal stated in Opinion **G 3/08** (OJ 2011, 10) that the case law of the boards of appeal as a whole is consistent in considering all the features that are claimed. The boards have always avoided approaches which involve weighting of features or a decision which features define the "essence" of the invention. It is true that the COMVIK/Hitachi (**T.641/00**, **T.258/03**) approach to deciding whether there is an inventive step may involve ignoring some features, but the method starts with a consideration of all the features together to determine whether the claimed subject-matter has a technical character. Only once this determination has been made can the board turn to the question of which claimed features contribute to that technical character and therefore should be taken into account for the assessment of whether there is an inventive step (see also **T.528/07**).

The Enlarged Board of Appeal went on and stated that it is in fact a well-established principle that features which would, taken in isolation, belong to the matters excluded from patentability by Art. 52(2) EPC may nonetheless contribute to the technical character of a claimed invention, and therefore cannot be discarded in the consideration of the inventive step. This principle was already laid down, albeit in the context of the so-called "contribution approach", in one of the earliest decisions of the boards of appeal to deal with Art. 52(2) EPC, namely **T.208/84**.

The second problem with the alleged divergence was that the decisions **T.163/85** and **T.190/94** said in the referral to require a technical effect on a physical entity in the real world, simply did not do so. They merely accepted this as something sufficient for avoiding exclusion from patentability; they did not state that it was necessary.

Dealing with this issue again in **G 1/19**, the Enlarged Board found as follows: beyond those considered in **G 3/08**, the referring board identified cases apparently requiring a technical effect directly linked to physical reality, but also others which suggested that a potential technical effect, i.e. an effect achieved only in combination with non-claimed features, was taken into account (**T.1351/04**, **T.625/11**, see points 36 and 37 of the reasons of **T.489/14** of 22 February 2019, OJ 2019, A86). Following existing case law and taking into account the relevant legal provisions, the Enlarged Board in **G 1/19** (OJ 2021, A77) did not see a need to require a direct link with (external) physical reality in every case. On the one hand, technical contributions might also be established by features within the computer system used. On the other hand, there were many examples in which potential technical effects - which could be distinguished from direct technical effects on physical reality - had been considered in the course of the technicality / inventive step analysis. While a direct link with physical reality, based on features that per se are technical and/or non-technical, was in most cases sufficient to establish technicality, it could not be a necessary condition, if only because the notion of technicality needs to remain open.

c) Potential technical effect

In G 1/19 the argument was put forward, that it was sufficient for a computer-implemented invention to have the potential to produce a technical effect, citing T 1173/97 as well as a number of other decisions (T 208/84, OJ 1987, 14; T 163/85, OJ 1990, 379; T 110/90, OJ 1994, 557; T 1351/04).

The Enlarged Board held that the principle developed in T 1173/97 that software (which in itself may only have "potential effects") is treated as software running on a computer is still applied, while the further analysis (i.e. whether the software causes further technical effects) is now carried out according to the COMVIK approach. When run on a computer, the combination of the claimed features must establish a technical invention. In the COMVIK analysis, the features have to be assessed as to their contribution to the technical character of the invention.

The Enlarged Board distinguished between the potential effects discussed in T 1173/97 and the potential technical effects seen in older cases such as T 208/84 and T 163/85. While the potential effects in T 1173/97 include all (technical and non-technical) effects resulting directly from the running of a program on a computer, i.e. effects occurring within the computer and relating to the hardware which executes the program, T 208/84 and T 163/85 appeared to confirm that data intended for controlling a technical device may be considered to have technical character because it has the potential to cause technical effects. These potential technical effects are "downstream" effects which may or may not be caused by said data output.

In the context of the problem-solution approach and the COMVIK approach, such potential technical effects may be considered if the data resulting from a claimed process is specifically adapted for the purposes of its intended technical use. In such cases, either the technical effect that would result from the intended use of the data could be considered "implied" by the claim, or the intended use of the data (i.e. the use in connection with a technical device) could be considered to extend across substantially the whole scope of the claimed data processing method.

Numerical data output from a computer is a necessary pre-condition for any effects that are caused, and the "downstream effects" can be seen as a potential effect of the software. However, the necessarily technical nature of some effects inside the computer does not mean that the "downstream" effects caused by the data output of the computer are necessarily of a technical nature.

d) Virtual or "calculated" technical effect

In G 1/19 it was argued that technical effects which are not achieved through an interaction with physical reality, but are calculated in such a way as to correspond closely to "real" technical effects or physical entities, should be treated as technical effects for the purposes of the COMVIK approach. In the Enlarged Board's view, virtual or calculated technical effects should be distinguished from potential technical effects which, for example when a computer program or a control signal for an image display device is put to its intended use,

necessarily become real technical effects. Calculated status information or physical properties concerning a physical object are information which may reflect properties possibly occurring in the real world. However, first and foremost, they are mere data which can be used in many different ways. There may exist exceptional cases in which such information has an implied technical use that can be the basis for an implied technical effect. Still, in general, data about a calculated technical effect is just data.

e) Tangible effect

The Enlarged Board in G 1/19 fully supported the view expressed in T 533/09 that a tangible effect is not a requirement under the EPC. Moreover, it was unclear to what extent the notions of "tangible effect" and "further technical effect" overlapped. A criterion based on tangibility - in addition to the requirement of technicality - thus cannot contribute to a more precise delimitation of patentable inventions.

f) Technical considerations: programming a computer

It is not specified in the EPC whether, or under which circumstances, the activity associated with creating programs for computers, i.e. programming a computer, is a technical activity which is in principle patentable, or a non-technical activity which is as such excluded from patentability.

The President of the EPO referred to the Enlarged Board of Appeal the following point of law (see G 3/08, OJ 2011, 10; Question 4 of the referral): Does the activity of programming a computer necessarily involve technical considerations? The President of the EPO cited decision T 1177/97, where the board found that "implementing a function on a computer system always involves, at least implicitly, technical considerations". The reference to 'computer routines' made it clear that this implementation consisted of programming the computer (see also T 172/03). Contrary to these decisions T 833/91, T 204/93 and T 769/92 (OJ 1995, 525) considered programming to be a mental act of the programmer.

In Opinion G 3/08 (OJ 2011, 10) the Enlarged Board of Appeal considered that the referral asserted correctly that T 1177/97 considered that programming always involved technical considerations, at least implicitly, and that T 172/03 assumed the same in that it considered the skilled person, who, it was emphasised, was a technical expert, to be a software project team, consisting of programmers. On the other hand, T 833/91, T 769/92 (OJ 1995, 525) and T 204/93, were said to consider the programmer's activity, programming, to be a mental act, falling within the exclusions of Art. 52(2) EPC.

The Enlarged Board stated that there was no contradiction between these positions, as may be seen by considering the same case in a non-controversial field, for example bicycle design. Designing a bicycle clearly involves technical considerations (it may also involve non-technical, e.g. aesthetic, considerations), but it is a process which at least initially can take place in the designer's mind, i.e. it can be a mental act and to the extent that it is a mental act would be excluded from patentability (cf. also T 833/91, T 769/92, T 204/93, T 914/02 and T 471/05). Hence the question did not satisfy the requirement for a divergence in the case law and was therefore inadmissible.